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10 CFR 50.73

January 20, 2022

Serial: RA-21-0344

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Brunswick Steam Electric Plant, Unit Nos. 1 and 2
Renewed Facility Operating License Nos. DPR-71 and DPR-62
Docket Nos. 50-325 and 50-324
Licensee Event Report 1-2021-001

In accordance with the Code of Federal Regulations, Title 10, Part 50.73, Duke Energy Progress, LLC, is submitting the enclosed Licensee Event Report (LER). This report fulfills the requirement for a written report within sixty (60) days of a reportable occurrence.

This document contains no regulatory commitments.

Please refer any questions regarding this submittal to Mr. Mark DeWire, Manager – Nuclear Support Services, at (910) 832-6641.

Sincerely,

A handwritten signature in black ink, appearing to read "John A. Krakuszeski".

John A. Krakuszeski

SBY/sby

Enclosure: ³ Licensee Event Report

cc (with enclosure):

Ms. Laura Dudes, NRC Regional Administrator, Region II
Mr. Luke Haeg, NRC Project Manager
Mr. Gale Smith, NRC Senior Resident Inspector
Chair - North Carolina Utilities Commission



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk alt: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Brunswick Steam Electric Plant (BSEP), Unit 1

2. Docket Number

05000325

3. Page

1 OF 3

4. Title

Automatic Specified System Actuation due to Loss of Power to Emergency Bus E3

5. Event Date

Month	Day	Year
12	06	2021

6. LER Number

Year	Sequential Number	Revision No.
2021	- 001 -	00

7. Report Date

Month	Day	Year
01	20	2022

8. Other Facilities Involved

Facility Name	Docket Number
Brunswick, Unit 2	05000324
Facility Name	Docket Number
	05000

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20

☐ 20.2203(a)(2)(vi)☐ 50.36(c)(2)☒ 50.73(a)(2)(iv)(A)☐ 50.73(a)(2)(x)☐ 20.2201(b)☐ 20.2203(a)(3)(i)☐ 50.46(a)(3)(ii)☐ 50.73(a)(2)(v)(A)

10 CFR Part 73

☐ 20.2201(d)☐ 20.2203(a)(3)(ii)☐ 50.69(g)☐ 50.73(a)(2)(v)(B)☐ 73.71(a)(4)☐ 20.2203(a)(1)☐ 20.2203(a)(4)☐ 50.73(a)(2)(i)(A)☐ 50.73(a)(2)(v)(C)☐ 73.71(a)(5)☐ 20.2203(a)(2)(i)

10 CFR Part 21

☐ 50.73(a)(2)(i)(B)☐ 50.73(a)(2)(v)(D)☐ 73.77(a)(1)(i)☐ 20.2203(a)(2)(ii)☐ 21.2(c)☐ 50.73(a)(2)(i)(C)☐ 50.73(a)(2)(vii)☐ 73.77(a)(2)(i)☐ 20.2203(a)(2)(iii)

10 CFR Part 50

☐ 50.73(a)(2)(ii)(A)☐ 50.73(a)(2)(viii)(A)☐ 73.77(a)(2)(ii)☐ 20.2203(a)(2)(iv)☐ 50.36(c)(1)(i)(A)☐ 50.73(a)(2)(ii)(B)☐ 50.73(a)(2)(viii)(B)☐ 20.2203(a)(2)(v)☐ 50.36(c)(1)(ii)(A)☐ 50.73(a)(2)(iii)☐ 50.73(a)(2)(ix)(A)☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Mark DeWire, Manager – Nuclear Support Services

Phone Number (Include area code)

(910) 832-6641

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

At 11:25 Eastern Standard Time (EST) on December 6, 2021, during a planned Emergency Diesel Generator (EDG) 3 maintenance outage, with Unit 1 and 2 in Mode 1 at approximately 100% power, electrical power was lost to 4160V emergency bus E-3. By design this resulted in automatic isolation of various Unit 2 Primary Containment Isolation System (PCIS) Group 2, 3, 6, and 10 isolation valves, and Unit 1 PCIS Group 6 isolation valves. EDG 3 received an automatic start signal by design but was under clearance for planned maintenance and therefore did not start.

Electrical power to emergency bus E-3 was lost due to an inadvertent undervoltage signal generated during a planned relay replacement as part of the EDG 3 maintenance outage. During removal of a support bar to facilitate relay replacement, a washer was released contacting terminals directly below the bolt being removed causing control power fuses to blow. This resulted in an undervoltage signal and the loss of electrical power to emergency bus E-3. The washer that was released was not on any design drawings and its location, behind the support bar, precluded viewing and establishment of preventative measures.

Emergency bus E-3 was re-energized on December 6, 2021, at 13:12 EST via its normal power source (i.e., Unit Auxiliary Transformer). There was no impact on the health and safety of the public or plant personnel. The safety significance of this event is minimal.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) due to actuation of the PCIS and EDG 3.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Brunswick Steam Electric Plant (BSEP), Unit 1	05000- 325	2021	- 001	- 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

Background*Initial Conditions*

At the time of the event, Unit 1 was in Mode 1 (i.e., Power Operation), at approximately 100 percent rated thermal power. Unit 2 was in Mode 1 (i.e., Power Operation), at approximately 100 percent rated thermal power.

Reportability Criteria

This event is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A) because it involved actuation of systems listed in 10 CFR 50.73(a)(2)(iv)(B). Specifically, the Unit 1 and 2 Primary Containment Isolation System (PCIS) [JM], and Emergency Diesel Generator (EDG) [EK] 3 actuated during this event.

The NRC was notified of this event per 10 CFR 50.72(b)(3)(iv)(A) via Event Notification 55627 at 18:14 Eastern Standard Time (EST) on December 6, 2021. An update was provided on December 6, 2021, at 20:28 EST.

Event Description

At 11:25 EST on December 6, 2021, during a planned maintenance outage on EDG 3, with Unit 1 and 2 in Mode 1 at approximately 100% power, electrical power was lost to 4160V emergency bus [EB] E-3. EDG 3 received an automatic start signal by design but was under clearance for planned maintenance and therefore did not start.

By design the loss of power to emergency bus E-3 and corresponding loads resulted in automatic isolation signals to various Unit 2 PCIS Group 2 (i.e., Drywell Equipment and Floor Drain, Residual Heat Removal (RHR) Discharge to Radwaste, RHR Process Sample, and Traversing Incore Probe), Group 3 (i.e., Reactor Water Cleanup), Group 6 (i.e., Containment Atmosphere Control/Dilution, Containment Atmosphere Monitoring, and Post Accident Sampling Systems), and Group 10 (i.e., Drywell Pneumatics) isolation valves. In addition, automatic isolation of the Unit 1 PCIS Group 6 isolation valves occurred due to the resulting loss of power to the Main Stack High Radiation logic which is powered from Unit 2. Appropriate primary containment isolation valves closed as expected.

Other Unit 1 and 2 actuations included Reactor Building Ventilation System [VA] isolation (i.e., Secondary Containment isolation) and automatic start signals to the Standby Gas Treatment System [BH] and Control Room Emergency Ventilation System [VI]. The affected equipment responded as designed.

Event Cause

As part of the planned EDG 3 maintenance outage, the EDG 3 Synchronization (Sync) Check Relay was being replaced. Inside the associated electrical compartment, a metal support bar connects the Sync Check Relay, the Over/Undervoltage Relay, and the Undervoltage Relay. Installation of the new Sync Check Relay required removal of the metal support bar to facilitate modification of the support bar to accommodate the slightly different bolting configuration of the new Sync Check Relay. During removal of the support bar, a washer was released contacting the Undervoltage Relay terminals directly below the bolt being removed causing control power fuses to blow. This resulted in an undervoltage signal to the protective relay logic which tripped the incoming breaker resulting in the loss of electrical power to emergency bus E-3.

The washer that was released was not on any design drawings and its location, behind the support bar, precluded viewing and establishing measures to prevent the event. Work history was reviewed following the event and no work has been performed on these relays which would have disturbed the relay cases or the associated support bar, indicating that this washer configuration was part of original plant construction.

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Brunswick Steam Electric Plant (BSEP), Unit 1	05000- 325	2021	- 001	- 00

NARRATIVE

The cause of this event was insufficient rigor in documenting the as-built configuration of the equipment with the washer located behind the support bar. This condition has existed since original plant construction and is considered a historical condition.

Safety Assessment

There was no impact on the health and safety of the public or plant personnel as a result of this event. The safety significance of this event is minimal. The automatic system actuations occurred as expected, and all safety related systems operated as designed.

Corrective Actions

Emergency bus E-3 was re-energized on December 6, 2021, at 13:12 EST via its normal power source (i.e., Unit Auxiliary Transformer [EA]) following replacement of the associated control power fuses. The modified relay support bar was reinstalled on December 7, 2021.

In addition, on December 16, 2021, signs were installed inside the Sync Check Relay cabinets associated with emergency bus E-3 and E-4 identifying the potential for washers to be present behind relay support bars. Also, to prevent recurrence during the remaining Sync Check Relay replacements associated with emergency bus E-1 and E-2, work orders were updated on January 20, 2022, to install signs inside the cabinets identifying the potential for washers to be present behind relay support bars, and to place insulating material behind the terminals of the adjacent relays prior to removing hardware from the relay support bar.

Any revisions to corrective actions will be made in accordance with the site's corrective action program.

Previous Similar Events

A review of events for the past three years identified the following previous similar events related to automatic system actuations during planned maintenance activities.

- Event Notification 54144, completed on July 3, 2019, reported an invalid actuation of EDG 1 during planned maintenance when the starting air clearance was being lifted while simultaneously performing a Post Maintenance Test (PMT) where an external DC power source was applied to a relay that provided continuity directly to the starting air solenoids resulting in the energization of the air start solenoids causing EDG 1 to start. The event resulted from insufficient risk recognition for a first time PMT and shortfalls in the scheduling process.
- Event Notification 54675, completed on April 22, 2020, reported an invalid actuation of PCIS Group 6 isolation valves during a planned refueling outage. The invalid actuation occurred when power was lost as a result of removal of the inboard isolation logic fuse during a planned clearance hang to support maintenance. The event resulted from inadequate procedure use and adherence in clearance development.

The corrective actions associated with these previous similar events could not have reasonably been expected to prevent the condition reported herein.

Commitments

No regulatory commitments are contained in this report.